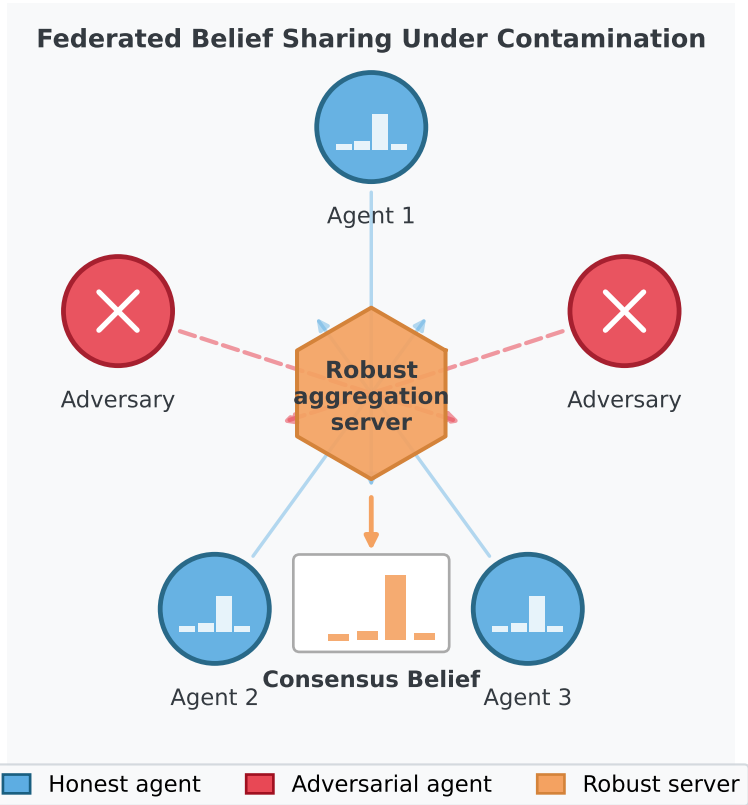


Federated belief sharing: from generative model to robust consensus

A categorical, recovery-tested bridge between active inference and generalized Bayes

Recovery anchor: $\text{robust_aggregate}(0) = \text{log_linear_pool} = \text{standard belief sharing}$

A



B

Consensus after contamination

Same broadcasts, different fusion rule

Naive log-linear pool

confident wrong broadcasts pull the product



39%

Heuristic reweighting

server suppresses outlying beliefs before fusion



50%

+11pp true-state mass

Deterministic schematic: 2/5 agents contaminated

Variational server: objective-backed fusion is a separate conservative rule

Three robustness axes — related routes, non-transferable guarantees

CLIENT-SIDE FEDGVI

source / β generalized-Bayes update
source-conditional bounded-influence result

SERVER HEURISTIC

robust_aggregate reweighting
conditional accuracy; recovery limit only

VARIATIONAL SERVER

variational_aggregate and $F(q,a)$
objective-backed raw-weight control; conservative